Ramsey El-Naggar

Software Engineer

Contact

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Languages

- English: Native
- Spanish: B1 (DELE 2021)

Skills

Main languages (and frameworks / versions) in order of proficiency:

1. Python (Django)
2. PHP
3. Java
4. SQL (MySQL, PostgreSQL)
5. HTML/CSS (Bootstrap)
6. Ruby (Ruby on Rails)
7. JavaScript

Additional languages/tools:

- Git, Bash, Docker, AWS
- Apache HTTP Server
- MATLAB, C, Haskell

Coursework

Compiling Techniques (100%)

Implemented (in Java) sections of a compiler for a subset of C, including: lexing, parsing, semantic analysis and Java bytecode generation.

System Design Project (88%)

Led a group of four to design and build a football-playing robot, placing second in the end-of-year competition, scoring six goals, and winning the prize for the best technical presentation in the year.

Experience

Computer Science and Programming Tutor

Self-employed | 2019 - present

• Designated the UK's top "Computing University" on the leading tuition platform MyTutor and received over 600 5-star ratings and reviews.
• Taught web development (HTML/CSS, Django, PHP, Ruby on Rails), object-oriented programming and software engineering (Python, Java), databases (MySQL), algorithms and data structures.

Full Stack Developer

N-able | 2018

• Front-end: upgraded plain JavaScript to Ext JS (a JS framework).
• Back-end: wrote a custom object-relational mapping (ORM) in PHP and SQL (MySQL) to uncouple the domain logic from the database layer, thus upgrading the procedural PHP to object-oriented PHP.
• Dev-ops/testing: configured Jasmine and PHPUnit within the codebase in order to use test-driven development (TDD) throughout the upgrade.

Education

Master of Informatics (MInf, CEng, CITP, CSci)

The University of Edinburgh | 2013 - 2018

• Awarded a degree classification of First Class Honours (*1st*).
• Dissertation (*82%*): Built a dung beetle robot with a neural model of navigational behaviour and awarded a grant to travel to South Africa and test the robot against real dung beetles.
• Achieved 83%-99% in all software engineering courses:

Data and Analysis (*99%*) | Elements of Programming Languages (*98%*) | Object-Oriented Programming (*97%*) | Computer Security (*92%*) | Computer Communication and Networking (*90%*) | Computer Systems (*90%*) | Algorithms, Data Structures and Learning (*86%*) | Software Engineering (*83%*)

Projects

<rmsy.me>: A personal website

rmsy.me/github

To showcase front- and back-end development skills.

• Built a reusable back-end PHP framework.
• Designed a responsive front-end with Bootstrap.
• Organised the application into standalone containers using Docker.
• Automated the DevOps pipeline using Bash scripts.
• Configured the AWS EC2 instance and Apache Server manually.

Extras

• Awarded the British Computer Society (BCS) Prize for Top Performance in Professional Issues (*99%*).
• Received an Edinburgh Award for leading peer-assisted learning sessions and recruiting, training and managing 12 more leaders.
• Nominated three times for the "Best Student Who Tutors Award" while teaching 11 groups across six different informatics courses.
• Placed in the semi-finals of the national competition Voice Festival UK while musical director at the Edinburgh University A Cappella Society.